Three-Dimensional Finite-Difference Time-Domain Simulation of Electromagnetic Scattering: Validation Against Cavity Modes, an Absorbing-Boundary Reference and Mie Theory

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Abstract

This report describes a finite-difference time-domain (FDTD) solver for Maxwell's equations in one, two and three dimensions, written in C++. The three-dimensional engine uses the Yee staggered grid, a convolutional perfectly matched layer (CPML) at the outer boundary and a total-field/scattered-field (TFSF) plane-wave source. It is tested against three independent references. The resonances of a closed metal cavity agree with the discrete Yee eigenfrequencies to 0.06 ppm, and their deviation from the continuum converges at second order (2.004 to 2.011 for eight modes). Compared with a reflection-free reference run, a 12-cell CPML gives a worst-case field error of -95.6 dB. The scattering efficiency of a dielectric sphere ($\varepsilon_r = 4$) agrees with Mie theory to a mean of 1.9% at 20 cells per radius and 1.3% at 30. The same solver then maps the focus of a graded-index sphere, which reaches $|E_z/E_0|^2 = 9.9$ inside the sphere near its rear surface.

1 Introduction

Analytic solutions of Maxwell's equations exist only for a few geometries. For anything else the fields have to be computed numerically. The finite-difference time-domain method, introduced by Yee in 1966 [1], solves the time-dependent curl equations directly on a staggered grid [2]. It needs no matrix inversion and handles broadband pulses in a single run.

A simulation is only useful if its errors are known. This project therefore builds the solver in stages (1D, 2D, then 3D) and checks the 3D engine against references that test one part of the method at a time:

- the Yee update itself, through the resonances of a closed metal cavity;
- the absorbing boundary, through a comparison with a much larger domain whose boundary the wave never reaches;
- the plane-wave source and the scattering calculation together, through Mie theory for a dielectric sphere.

With those checks in place, the solver is used on a problem without a closed form: the focusing of a graded-index (GRIN) sphere.

2 Maxwell's Equations

In a linear, non-magnetic, source-free medium the curl equations are

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}, \tag{1}$$

$$\nabla \times \mathbf{H} = \varepsilon_0 \varepsilon_r \frac{\partial \mathbf{E}}{\partial t}, \tag{2}$$

where $\varepsilon_r(\mathbf{r})$ is the relative permittivity. In a uniform dielectric the phase velocity is $v = c/\sqrt{\varepsilon_r}$, so the wavelength inside the medium shrinks by the same factor.

3 The Yee Scheme

3.1 Staggered grid

Space is divided into cubic cells of side Δ, and time into steps Δt. Each field component is stored half a cell away from its neighbours (Fig. 1):

$$E_x(i+\tfrac{1}{2}, j, k), \quad E_y(i, j+\tfrac{1}{2}, k), \quad E_z(i, j, k+\tfrac{1}{2}),$$

$$H_x(i, j+\tfrac{1}{2}, k+\tfrac{1}{2}), \quad H_y(i+\tfrac{1}{2}, j, k+\tfrac{1}{2}), \quad H_z(i+\tfrac{1}{2}, j+\tfrac{1}{2}, k).$$

Every E component is then surrounded by the four H components that form its curl, and vice versa. In time the fields leapfrog: $\mathbf{E}$ is known at integer steps n and $\mathbf{H}$ at half steps $n + \frac{1}{2}$. All derivatives become central differences over one cell or one step, so the scheme is second-order accurate in space and time.

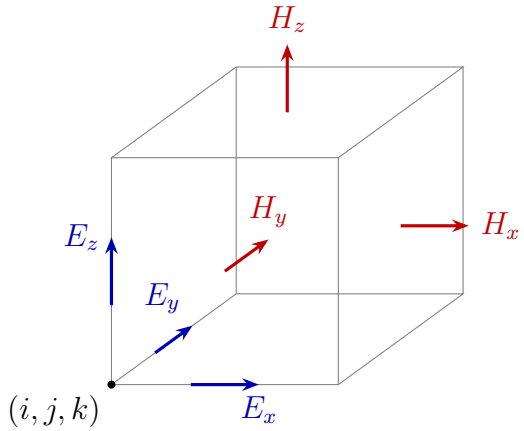


Figure 1: One Yee cell. Electric components (blue) sit at the centres of the cell edges and point along them; magnetic components (red) sit at the centres of the faces and point normal to them.

3.2 Update equations

With $c_E = \Delta t/(\varepsilon_0 \varepsilon_r \Delta)$ and $c_H = \Delta t/(\mu_0 \Delta)$, the x components update as

$$H_x^{n+\frac{1}{2}}(i, j+\tfrac{1}{2}, k+\tfrac{1}{2}) = H_x^{n-\frac{1}{2}} - c_H \Big[E_z^n(i, j+1, k+\tfrac{1}{2}) - E_z^n(i, j, k+\tfrac{1}{2}) - E_y^n(i, j+\tfrac{1}{2}, k+1) + E_y^n(i, j+\tfrac{1}{2}, k) \Big], \tag{3}$$

$$E_x^{n+1}(i+\tfrac{1}{2}, j, k) = E_x^n + c_E \Big[H_z^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j+\tfrac{1}{2}, k) - H_z^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j-\tfrac{1}{2}, k) - H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j, k+\tfrac{1}{2}) + H_y^{n+\frac{1}{2}}(i+\tfrac{1}{2}, j, k-\tfrac{1}{2}) \Big], \tag{4}$$

and the other four components follow by cyclic permutation of (x, y, z). The tangential $\mathbf{E}$ on the outermost faces is never updated, which makes the outer wall a perfect electric conductor (PEC). The 3D engine (`fdtd3d.hpp`) parallelises each update loop with OpenMP.

3.3 Stability

An explicit scheme is stable only if a wave cannot cross more than one cell per step. For the Yee scheme this is the Courant–Friedrichs–Lewy (CFL) condition

$$\Delta t \le \frac{1}{c\sqrt{1/\Delta x^2 + 1/\Delta y^2 + 1/\Delta z^2}}.$$

Table 1 lists the time steps used in this project.

Dimension	Limit	Used here
1D	$\Delta t \le \Delta z/c$	$\Delta t = \Delta z/(2c)$, $S = 0.5$
2D	$\Delta t \le \Delta/(c\sqrt{2})$	$0.99\times$ limit, $S = 0.70$
3D	$\Delta t \le \Delta/(c\sqrt{3})$	$0.99\times$ limit, $S = 0.5716$

Table 1: Time steps. The Courant number is $S = c\Delta t/\Delta$.

3.4 Numerical dispersion

On the grid a plane wave $e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$ satisfies

$$\frac{1}{(c\Delta t)^2}\sin^2\frac{\omega \Delta t}{2} = \sum_{u=x,y,z} \frac{1}{\Delta u^2}\sin^2\frac{k_u \Delta u}{2},$$

not $\omega = c|\mathbf{k}|$. Grid waves therefore travel slightly slower than c, with an error that falls as $(\Delta/\lambda)^2$. For the 3D runs at 75 GHz there are 57 cells per wavelength along an axis, and the phase-velocity error is 0.034%. Section 6 measures this dispersion directly.

4 One- and Two-Dimensional Demonstrations

4.1 1D: a Gaussian pulse in free space

The 1D code propagates a Gaussian pulse (E_x, H_y) on 200 cells of 100 μm for 1000 steps at $S = 0.5$. The source sits at cell 50. At $S = 0.5$ a wave crosses one cell in exactly two steps, so the boundary condition $E_0^{n+1} = E_1^{n-1}$ absorbs the outgoing wave exactly.

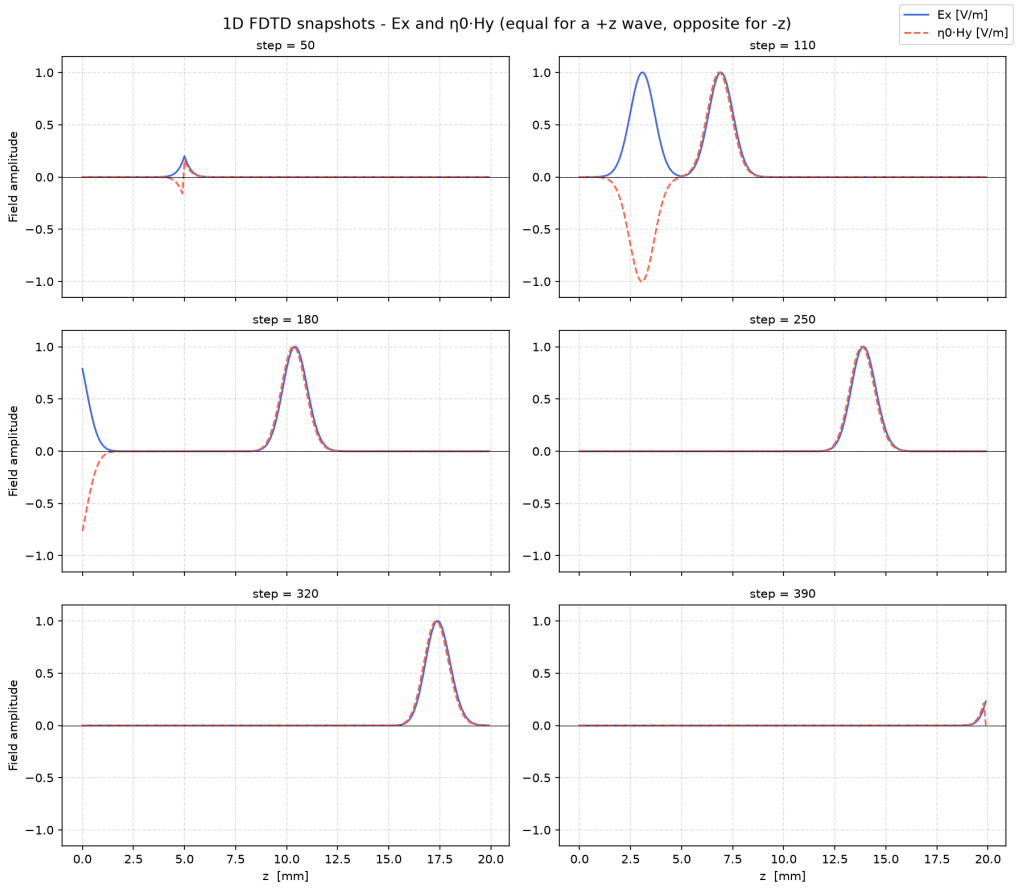


Figure 2: 1D snapshots of E_x and $\eta_0 H_y$. The source launches two pulses. In the $+z$ pulse the two curves coincide and in the $-z$ pulse they have opposite signs, as required for $\mathbf{E} \times \mathbf{H}$ to point along the direction of travel. Both pulses leave through the boundaries without reflecting.

The ratio E_x/H_y in the $+z$ pulse is 378 Ω. The 0.4% difference from $\eta_0 = 376.7\,\Omega$ comes from averaging the staggered H_y onto the E_x positions.

4.2 2D: a continuous line source

The 2D code solves the TM$_z$ equations (E_z, H_x, H_y) on 200 $\times$ 200 cells of 100 μm. A soft source at the centre radiates at 75 GHz ($\lambda = 4$ mm $= 40$ cells), switched on with a 200-step cosine ramp to avoid a broadband transient. The outer boundary is a first-order Mur condition [7].

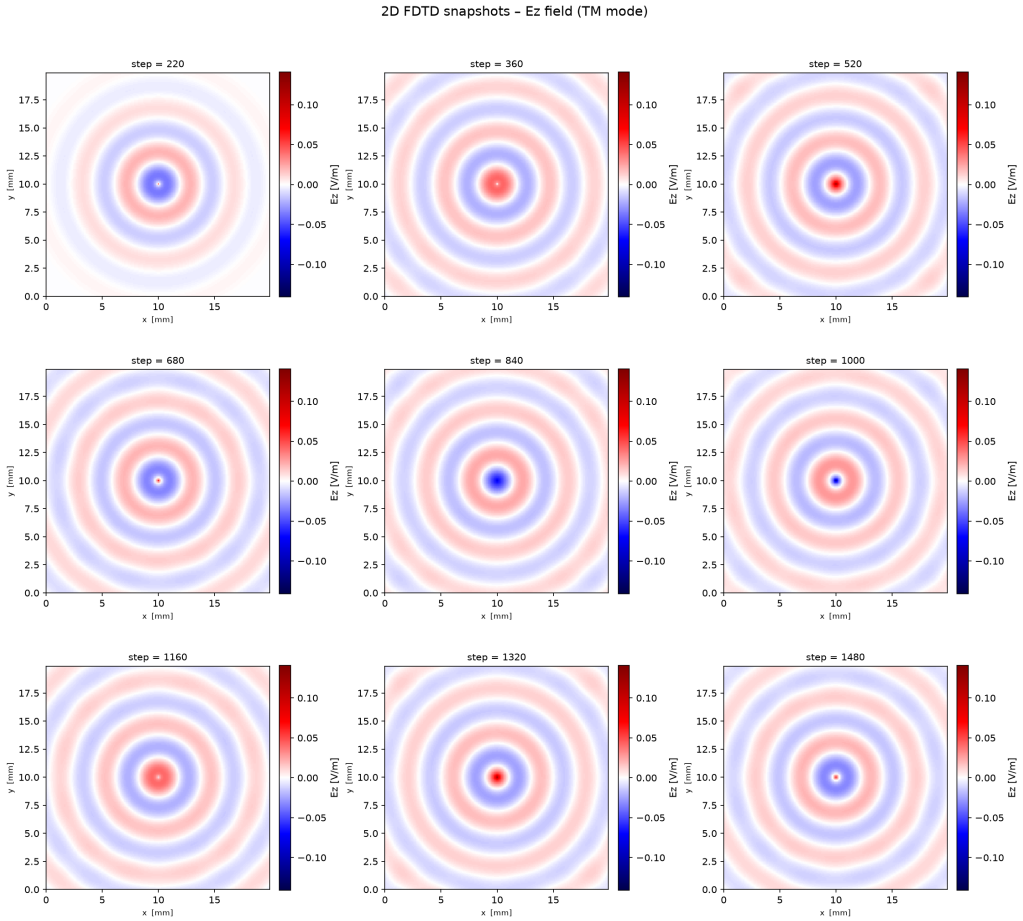


Figure 3: E_z of the 2D line source between steps 220 and 1480. The cylindrical wavefronts stay circular across the grid. Faint distortion near the corners comes from the first-order Mur boundary, which reflects a few percent at oblique incidence.

The 2D run is a qualitative demonstration. All quantitative tests below use the 3D engine.

5 The Three-Dimensional Engine

5.1 Convolutional perfectly matched layer

The domain is surrounded by a CPML [4, 3]: a layer in which the coordinate is stretched as

$$s_u = \kappa_u + \frac{\sigma_u}{\alpha_u + i\omega\varepsilon_0},$$

so that outgoing waves decay without reflecting at the interface. The profiles are graded with depth $d \in [0, 1]$ into the layer:

$$\sigma(d) = \sigma_{\max} d^m, \qquad \kappa(d) = 1 + (\kappa_{\max} - 1) d^m, \qquad \alpha(d) = \alpha_{\max}(1 - d),$$

with $m = 3$. In the time domain each stretched derivative needs an auxiliary field ψ, updated recursively with

$$b = e^{-(\sigma/\kappa+\alpha)\Delta t/\varepsilon_0}, \qquad a = \frac{\sigma}{\kappa(\sigma+\kappa\alpha)}(b-1).$$

The ψ arrays are stored only inside the layer of their own axis. The textbook optimum for the conductivity is [2]

$$\sigma_{\mathrm{opt}} = \frac{0.8(m+1)}{\eta_0 \Delta}.$$

Section 7 chooses $\sigma_{\max}$ and $\kappa_{\max}$ from a scan. The production runs use 12 cells with $\sigma_{\max} = 0.75\,\sigma_{\mathrm{opt}}$, $\kappa_{\max} = 1$ and $\alpha_{\max}/\varepsilon_0 = 10^{11}\,\mathrm{s}^{-1}$.

5.2 Material averaging

A sphere on a cubic grid has a staircase surface. To soften this, each E component takes the average ε_r over a $4 \times 4 \times 4$ sub-grid of the cell around its own staggered position. This sub-pixel averaging reduces, but does not remove, the staircase error (Section 11).

5.3 Total-field/scattered-field source

The incident wave is a plane wave travelling along $+x$ with E_z polarisation. A rectangular TFSF box splits the grid: inside, the code updates the total field; outside, only the scattered field. On the six faces of the box the update equations are corrected by the incident field, which comes from a separate 1D Yee grid with the same Δ and Δt [6]. For incidence along an axis the 1D grid has exactly the same numerical dispersion as the 3D one, so the correction cancels the incident wave outside the box to round-off.

The source pulse is a Gaussian-modulated sinusoid centred at 70 GHz, whose spectrum covers 30 to 110 GHz above e^{-2} of its peak.

5.4 Scattering efficiency

A closed box just outside the TFSF box records the tangential scattered fields. Their discrete Fourier transforms are accumulated during the run at 41 frequencies from 30 to 110 GHz. The scattered power at each frequency is the outward Poynting flux

$$P_{\mathrm{sca}} = \tfrac{1}{2}\,\mathrm{Re} \oint (\mathbf{E} \times \mathbf{H}^{*}) \cdot \hat{\mathbf{n}}\, dA,$$

and the scattering efficiency is

$$Q_{\mathrm{sca}} = \frac{P_{\mathrm{sca}}}{I_{\mathrm{inc}}\, \pi a^2},$$

with I_{inc} the incident intensity from the 1D grid at the sphere centre. The run continues until the field on the box has fallen to 10^{-6} of its peak, so the transforms are not truncated. Mie theory for the same sphere follows Bohren and Huffman [8].

6 Validation 1: Numerical Dispersion in a PEC Cavity

A closed PEC box of $12 \times 14 \times 17$ cells ($\Delta = 70\,\mu$m) has resonances at known frequencies. On the grid, the dispersion relation of Section 3.4 gives the *discrete Yee* eigenfrequencies exactly. In the continuum they are $f = \frac{c}{2}\sqrt{(m/L_x)^2 + (n/L_y)^2 + (p/L_z)^2}$. A broadband E_z pulse excites the cavity, a probe records 32 768 steps, and the peaks of the Hann-windowed spectrum give the measured frequencies. The run is repeated at $\Delta/2$ ($24 \times 28 \times 34$ cells, 65 536 steps).

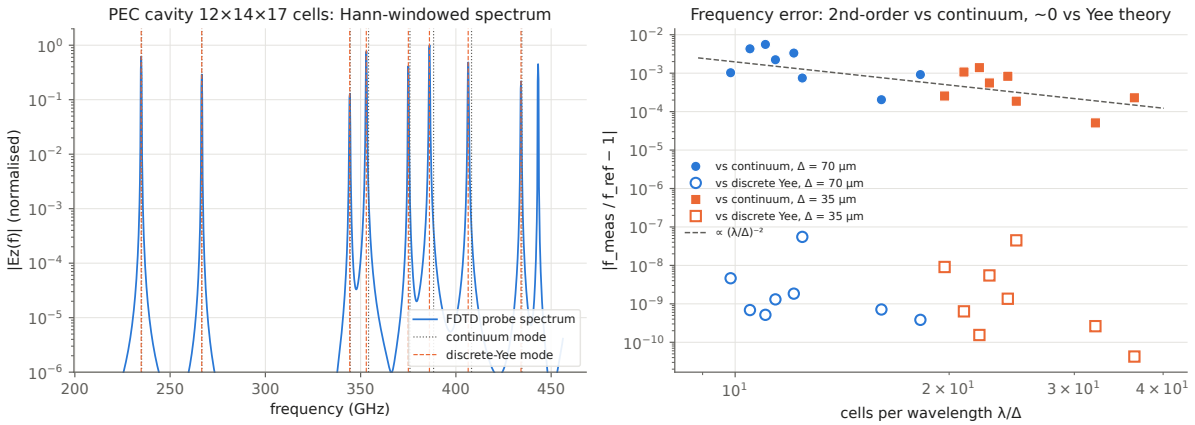


Figure 4: Left: probe spectrum of the cavity with the discrete-Yee (dashed) and continuum (dotted) eigenfrequencies. Right: relative frequency error against cells per wavelength. Against the continuum the error falls as $(\lambda/\Delta)^{-2}$; against the discrete Yee frequencies it is 10^{-10} to 10^{-7} at both resolutions.

Mode (m,n,p)	f_{cont} (GHz)	λ/Δ (Δ)	vs Yee (ppm) Δ	vs Yee (ppm) $\Delta/2$	vs continuum (%) Δ	vs continuum (%) $\Delta/2$	Order
(1,1,0)	235.030	18.2	0.000	0.000	0.092	0.023	2.004
(1,1,1)	266.657	16.1	0.001	0.000	0.020	0.005	2.004
(1,1,2)	344.537	12.4	0.055	0.045	0.075	0.019	2.006
(1,2,0)	354.154	12.1	0.002	0.001	0.333	0.083	2.007
(1,2,1)	375.888	11.4	0.001	0.005	0.224	0.056	2.007
(2,1,0)	388.291	11.0	0.001	0.000	0.560	0.139	2.008
(2,1,1)	408.212	10.5	0.001	0.001	0.431	0.107	2.008
(1,2,2)	434.617	9.9	0.005	0.009	0.103	0.025	2.011

Table 2: Cavity resonances. "Order" is $\log_2$ of the ratio of the continuum errors at Δ and $\Delta/2$.

The measured frequencies match the discrete Yee eigenfrequencies to at most 0.06 ppm. The update equations are therefore correct, and every difference from continuum physics is the predicted grid dispersion. That difference ranges from 0.020% to 0.56% at $\Delta = 70\,\mu$m and converges with orders between 2.004 and 2.011, as expected for a second-order scheme.

7 Validation 2: the Absorbing Boundary

A differentiated-Gaussian dipole sits at the centre of a 30-cell vacuum interior surrounded by CPML. Probes record E_z two cells inside the CPML interface in the face, edge and corner directions. The reference is the same source in a 218^3-cell domain, large enough that its own boundary cannot reach the probes within the 300-step window. The error at each probe is

$$\text{error} = \frac{\max_t |E - E_{\text{ref}}|}{\max_t |E_{\text{ref}}|},$$

and the table reports the worst of the three probes.

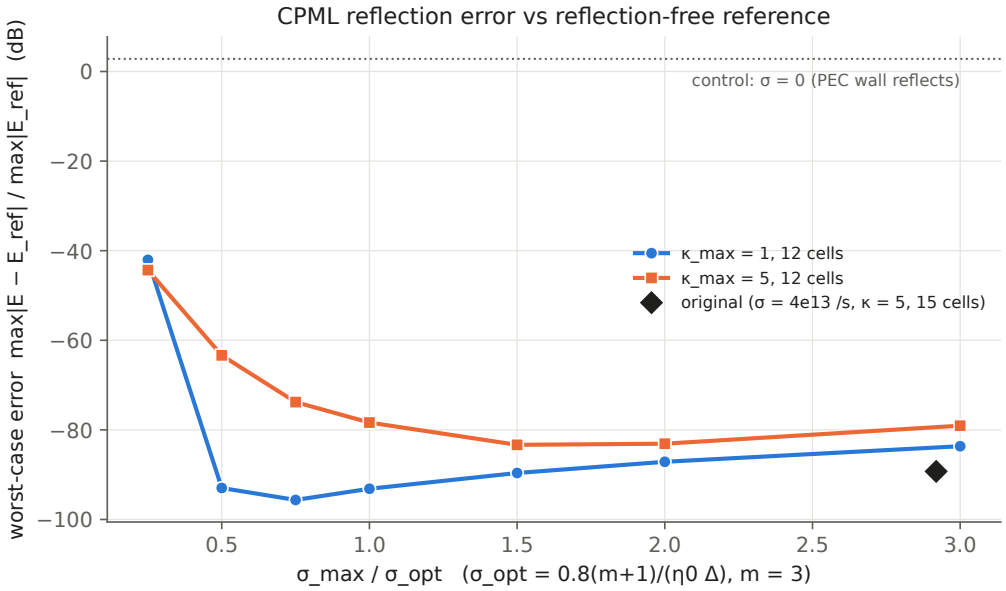


Figure 5: Worst-case CPML error against $\sigma_{\max}/\sigma_{\text{opt}}$ for 12-cell layers with $\kappa_{\max} = 1$ and 5. The diamond is the 15-cell layer with the original parameters; the dotted line is the control with $\sigma = 0$, where the PEC wall behind the layer reflects everything.

CPML	Worst-case error
12 cells, $\sigma_{\max} = 0.75\,\sigma_{\text{opt}}$, $\kappa_{\max} = 1$ (used)	-95.6 dB
12 cells, $\sigma_{\max} = \sigma_{\text{opt}}$, $\kappa_{\max} = 1$	-93.1 dB
12 cells, $\sigma_{\max} = 0.25\,\sigma_{\text{opt}}$, $\kappa_{\max} = 1$	-42.0 dB
15 cells, $\sigma_{\max} = 2.9\,\sigma_{\text{opt}}$, $\kappa_{\max} = 5$ (original)	-89.3 dB
control: $\sigma = 0$	$+2.8$ dB

Table 3: CPML error against a reflection-free reference.

Too little conductivity lets the wave reach the PEC wall and return (-42 dB at $0.25\,\sigma_{\text{opt}}$). Too much makes the grading too steep and reflects at the layer itself. The best layer sits slightly below σ_{opt}. For this broadband source $\kappa_{\max} = 5$ is worse at every conductivity from $0.5\,\sigma_{\text{opt}}$ upwards. A 12-cell layer with the scanned parameters beats the original 15-cell one by 6 dB while using fewer cells.

These numbers are field errors at probes near the boundary for a broadband pulse, which is the usual CPML benchmark. They are not plane-wave reflection coefficients.

8 Validation 3: Scattering by a Dielectric Sphere

8.1 Set-up

A homogeneous sphere with $\varepsilon_r = 4$ and radius $a = 1.4$ mm is illuminated by the TFSF plane wave. Across 30 to 110 GHz its size parameter $x = ka$ runs from 0.88 to 3.2, which covers the first resonances. The radius is resolved by $R = 10$, 20 and 30 cells, so that $\Delta = a/R$ and the grid grows from 68^3 to 108^3 cells.

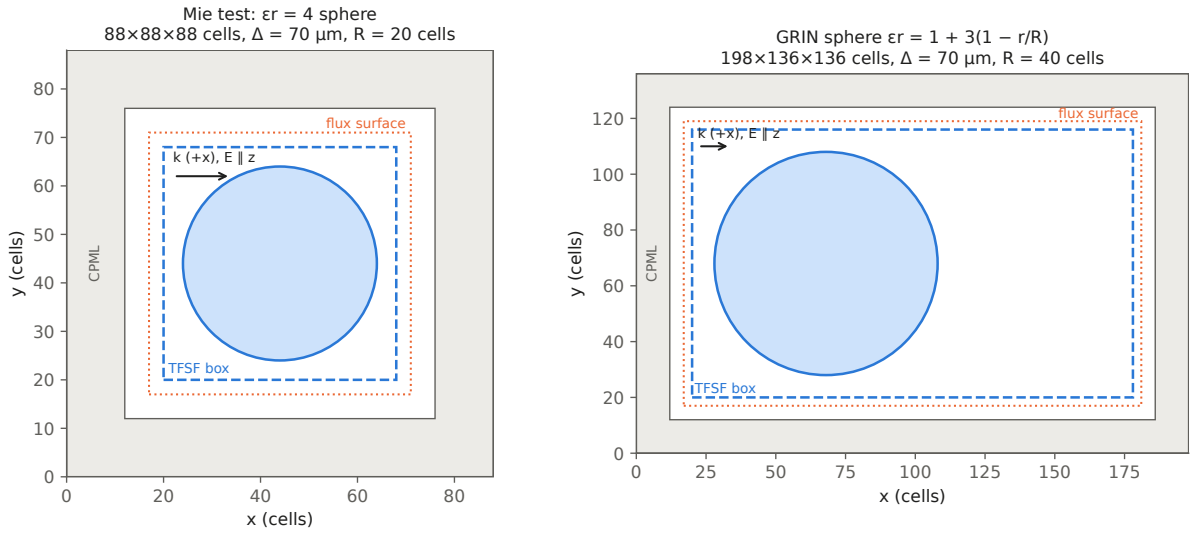


Figure 6: Simulation geometry in the plane $z =$ centre. Left: the Mie test (88^3 cells, $R = 20$). Right: the GRIN sphere ($198 \times 136 \times 136$ cells, $R = 40$), with extra total-field room behind the sphere for the focus. The plane wave travels along $+x$ with $\mathbf{E} \parallel \hat{z}$.

8.2 Results

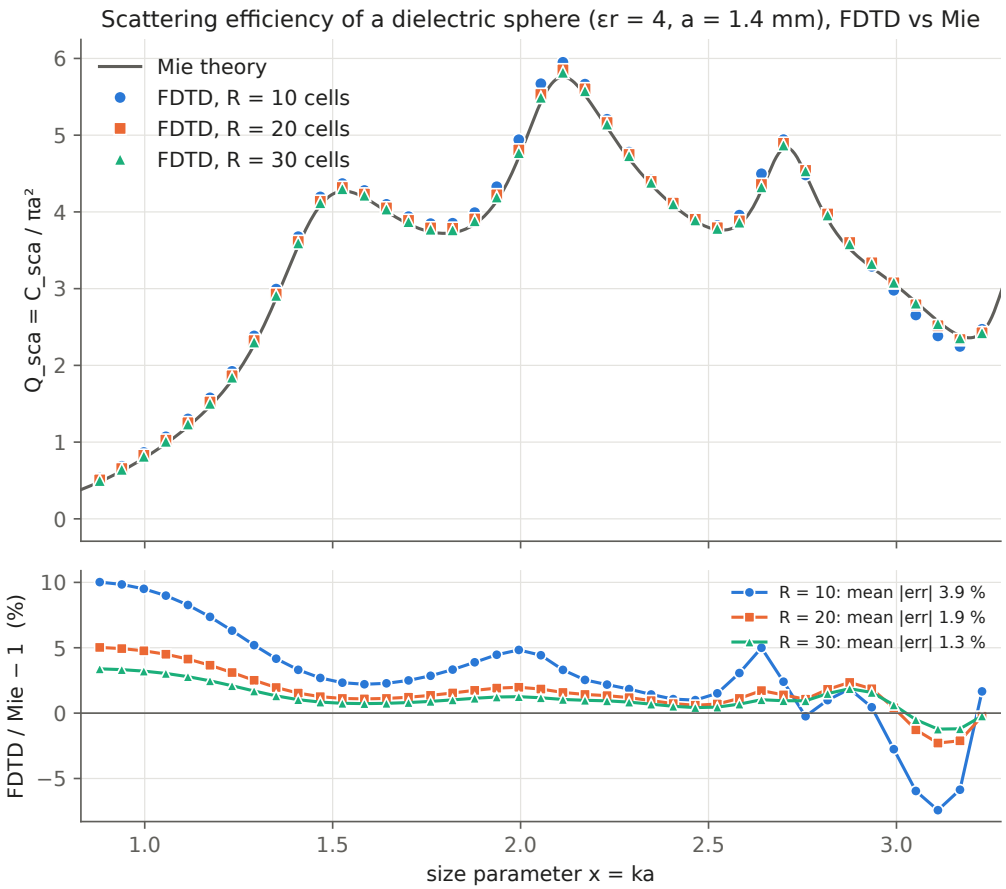


Figure 7: Top: scattering efficiency from FDTD at three resolutions against Mie theory. Bottom: relative error. The largest errors appear at small x, where the scattered power is weakest, and near the resonance at $x \approx 3.1$.

R (cells)	Δ	Grid	Mean $\lvert Q_{\mathrm{FDTD}}/Q_{\mathrm{Mie}} - 1\rvert$	Max
10	140 μm	68^3	3.93%	10.0%
20	70 μm	88^3	1.91%	5.0%
30	46.7 μm	108^3	1.30%	3.4%

Table 4: Scattering efficiency against Mie theory over 41 frequencies.

FDTD follows every peak and dip of the Mie curve. The error halves when the resolution doubles: it falls roughly as Δ, which is first order. This is slower than the second-order convergence of the cavity test, and the difference comes from the staircased sphere surface, where sub-pixel averaging only partly restores accuracy. The FDTD values sit slightly above Mie at small x at every resolution.

With the sphere removed, the field on the scattering surface should be zero. The measured leakage is 1.2×10^{-15} of the incident field (-299 dB), which is round-off. The TFSF correction therefore adds no spurious scattered power.

9 Focusing by a Graded-Index Sphere

The validated solver is now applied to a sphere whose permittivity falls linearly from the centre to the surface:

$$\varepsilon_r(r) = 1 + 3\left(1 - \frac{r}{R}\right), \qquad r < R,$$

with $R = 40$ cells $= 2.8$ mm, which is 0.7λ at 75 GHz. The grid has $198 \times 136 \times 136$ cells of $70\,\mu$m, with 70 extra cells behind the sphere so that the focus lies inside the total-field region.

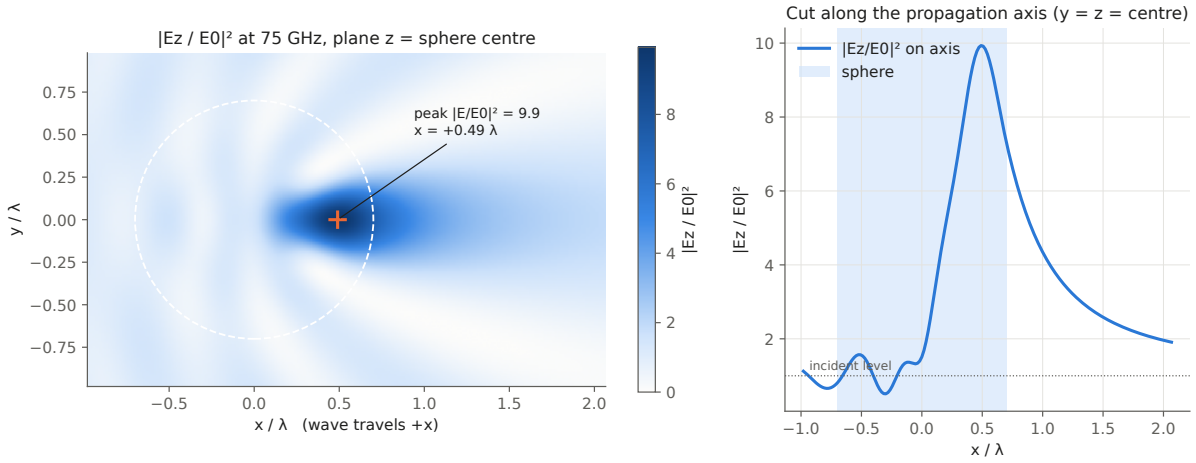


Figure 8: Left: $|E_z/E_0|^2$ at 75 GHz in the plane through the sphere centre; the dashed circle is the sphere surface. Right: the same quantity along the propagation axis, with the sphere shaded.

The intensity peaks at $|E_z/E_0|^2 = 9.9$ on the propagation axis, $0.70R$ (0.49λ) behind the centre, where $\varepsilon_r = 1.9$. The focus therefore lies inside the sphere, near its rear surface. Behind the sphere the intensity decays slowly along the axis and stays above twice the incident level for more than a wavelength. In front of the sphere the incident and reflected waves form a weak standing pattern. There is no analytic reference for this geometry, so the result is a demonstration of the solver rather than a test.

10 Simulation Parameters

Parameter	1D	2D	3D
Cell size Δ	$100\,\mu$m	$100\,\mu$m	$70\,\mu$m (Mie: 1.4 mm$/R$)
Courant number S	0.5	0.70	0.5716
Grid	200	200×200	88^3 (Mie, $R = 20$); $198 \times 136 \times 136$ (GRIN)
Source	Gaussian pulse	75 GHz CW, cosine ramp	TFSF Gaussian-modulated pulse
Boundary	exact two-step ABC	first-order Mur	CPML, 12 cells

Table 5: Main parameters of the three solvers.

The 3D engine stores six field components and three permittivity coefficients per node, 72 bytes per node, plus ψ arrays inside the CPML slabs only.

11 Limitations

- The sphere surfaces are staircased. Sub-pixel averaging of ε_r offsets this only partly, and it is the main source of the Mie error, which falls as Δ rather than Δ^2.
- The TFSF source supports incidence along a grid axis only. Oblique incidence would need a 1D auxiliary grid matched to the 3D dispersion in that direction.
- The first-order Mur boundary of the 2D code reflects a few percent at oblique incidence, so the 2D results are qualitative.
- Materials are non-dispersive and isotropic.

12 Conclusion

The Yee update reproduces the discrete eigenfrequencies of a PEC cavity to 0.06 ppm and converges to the continuum at second order. A 12-cell CPML with $\sigma_{\max} = 0.75\,\sigma_{\text{opt}}$ keeps the boundary error at -95.6 dB. The TFSF source leaks at round-off level, and the scattering efficiency of a dielectric sphere matches Mie theory to 1.3% on average at 30 cells per radius, limited by the staircased surface. With these checks passed, the solver shows that a linear GRIN sphere of radius 0.7λ focuses a plane wave to $|E_z/E_0|^2 = 9.9$ just inside its rear surface.

Natural next steps are dispersive materials, conformal or anisotropic averaging at curved surfaces to recover second-order convergence, and oblique TFSF incidence.

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